



Team Details

Team Name:

perceptron

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	Aman Kumar Maurya	4th Year

i A team can have up to 4 members including the team leader. Add rows if necessary.

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Problem Statement Addressed



AI-Based Restoration of Degraded Images for Semiconductor Inspection (KLA)

DESCRIPTION / DETAILS

Inspection images arrive degraded by speckle noise, additive Gaussian noise and downsampling, applied in an undisclosed order. Input is half resolution and noisy; the output must be clean at full ground-truth resolution. Lost detail reduces the reliability of every downstream measurement and defect-detection step. Doing nothing (bicubic upsampling) scores 22.94 dB PSNR, and the hidden test set includes out-of-distribution content, so the model must generalise beyond the training images.



Proposed Solution – Describe your Idea/Solution/Prototype



Idea and methodology: recover the degradation model from the data, then invert it with a single efficient network.

SOLUTION DETAILS
KEY CONCEPT - Restoration is an inverse problem, so we first recovered the forward model by fitting the residual across all 3200 provided pairs: 2x2 area downsampling, noise applied after downsampling, and multiplicative Gamma speckle (measured skew +0.357 vs Gamma's predicted +0.333).

SOLUTION - A NAFNet-style network, 16 blocks running at LOW resolution with a single PixelShuffle x2 upsample, doing joint denoising and super-resolution in one forward pass. 0.68 M parameters, no BatchNorm, fully convolutional so the same weights serve 128→256 and 256→512.

TRAINING - 50% real pairs, 50% synthetic pairs generated from the measured distribution with randomised kernel and degradation order. Loss = Charbonnier + 0.15 x (1 - SSIM), plus LPIPS for the final 10% of steps.



Innovation and Uniqueness



Highlight what makes your idea unique or innovative compared to existing solutions.

KEY INNOVATION

We measured the degradation instead of guessing it. Most entrants feed the provided pairs to a network; we recovered the forward model first.

The residual's skew identifies the noise FAMILY: multiplicative Gaussian is symmetric, Gamma is not. Measured +0.357 against a predicted +0.333 pins it.

Synthetic pairs sampled from empirical quantile tables match reality; a first version using assumed ranges was 50% noisier and would have taught the model to over-smooth.

Leakage-aware validation: samples are ordered by source image, so we hold out 10 evenly-spaced contiguous blocks rather than a random split.

Deliberately no GAN - the brief requires restoration without hallucinating structure, and invented detail is harmful in a measurement context.



Impact and Benefits



Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

Primary Impact

Recovers detail lost to downsampling without inventing structure, which is what a precision-measurement pipeline requires.
Improves all three scored axes at once – perceptual metrics often move against PSNR, and here they do not.

Quantifiable Outcomes

PSNR 27.54 dB vs 22.94 dB bicubic (+4.60 dB)
SSIM 0.7594 vs 0.5460 (+39%)
LPIPS 0.3922 vs 0.4478 (lower is better)
Inference 21.7 ms/image (46 img/s on an NVIDIA T4)



Technology & Feasibility/Methodology Used



Technical stack, feasibility, and the research the approach rests on.

IMPLEMENTATION STRATEGY

STACK - PyTorch 2.x / Python 3.11. NAFNet-style CNN, 0.68 M parameters, trained from scratch on a single NVIDIA T4 within one free Kaggle session. Inference imports only torch and numpy. no internet, API keys, downloads or manual configuration.

RUN - python run.py <input-dir> <output-dir>. Weights auto-load from models/, the output directory is created if absent, outputs are clamped to [0,1] with no NaN/Inf. Verified end-to-end from a fresh public clone.

HYGIENE - Seeds fixed, config-driven, every validation logs metrics plus git hash; a pre-flight gate blocks a broken run before it wastes GPU hours.

REFERENCES - NAFNet (Chen 2022); Real-ESRGAN degradation modelling (Wang 2021); PixelShuffle (Shi 2016); SSIM (Wang 2004); LPIPS (Zhang 2018). No external datasets or pretrained restoration weights; lpips (BSD-2) supplies the perceptual metric only.



Restoration network



NVIDIA GPU (T4 / H100)



PyTorch, NumPy



GitHub & Video Link



GitHub Repository

<https://github.com/23f2001033/kla-restore>



Prototype / Simulation Video



Working solution is reproducible directly from the repository: `python run.py <input-dir> <output-dir>`